Sean Richards

Research Fellow - Department of Physics

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Education

Doctor of Philosophy (PhD) — Auckland, New Zealand
THE UNIVERSITY OF AUCKLAND — September 2021 - August 2025

- Supervisors: Prof. Jan (JJ) Eldridge & Assoc. Prof. Nicholas Rattenbury
- Thesis: *The Next Generation: Refining the Standard Population Synthesis Model to Better Predict Stellar Population Properties*

Master of Science (MSc) — Auckland, New Zealand
THE UNIVERSITY OF AUCKLAND — March 2020 - March 2021

- Supervisor: Prof. Jan (JJ) Eldridge
- Thesis: *The Missing Mergers: Modelling the observed binary neutron-star population and their mergers*
- Awarded with Honours (First Class)

Postgraduate Diploma in Science (PGDipSci) — Auckland, New Zealand
THE UNIVERSITY OF AUCKLAND — March 2019 - February 2020

- Supervisor: Assoc. Prof. Nicholas Rattenbury
- Dissertation: *The University of Auckland Ground Station One*
- Technical report: *The University of Auckland Ground Station One: System Overview*
- Technical report: *Ground Segment Support for APSS-I* (contains non-public information, and cannot be distributed)

Bachelor of Science (BSc) — Auckland, New Zealand
THE UNIVERSITY OF AUCKLAND — February 2016 - November 2018

- Majored in Physics and Computer Science

Professional Experience

2025-2026 Technician
The University of Auckland

Awards, Fellowships, & Grants

2026-2028 Research Fellow
The University of Auckland

2021-2025 University of Auckland Doctoral Scholarship
The University of Auckland

2019 First in Course Award (PHYSICS 746: Quantum Field Theory)
The University of Auckland

Presentations

** presenting author;*

Invited Talks

Richards, S, 2026. *The Next Generation: Refining the Standard Population Synthesis Model to Better Predict Stellar Population Properties*. Colloquium presentation: The University of Auckland Department of Physics, Auckland, New Zealand.

Richards, S, Burn, L. 2024. *Beyond the Bias: Student insights on equity from Auckland to Exeter.* Colloquium presentation: The University of Auckland Department of Physics, Auckland, New Zealand.

Contributed Presentations

Richards, S, 2026. *Binary stars through cosmic time.* Oral presentation: The 3rd Faculty of Science Pacific Research Symposium, Auckland, New Zealand.

Richards, S*, Eldridge, J, Ghodla, S, Briel, M. 2026. *The evolution of accretor stars in binary systems due to accretion of increasingly helium-rich material.* Oral presentation: National Astronomy Meeting 2026, Birmingham, United Kingdom.

Richards, S, 2024. *A fate written in stardust: Probing the interiors of interacting binary stars.* Oral presentation: The 2nd Faculty of Science Pacific Research Symposium, Auckland, New Zealand.

Richards, S*, Eldridge, J, Briel, M, Ghodla, S. 2024. *The variable evolution of accretor stars in binary systems due to accretion of increasingly helium-rich material.* Oral presentation: Annual Scientific Meeting of the Astronomical Society of Australia (online).

Richards, S*. 2022. *New constraints on the Bray conservation-of-momentum natal kick model from multiple distinct observations.* Oral presentation: Supernovae in the Gravitational Wave Detection Era, Melbourne, Victoria, Australia.

Teaching Experience

2026	PHYSICS 753 Course Coordinator	The University of Auckland
2023-2025	PHYSICS 102 Tutor	The University of Auckland
2021-2023	Tuākana Programme Tutor	The University of Auckland
2018-2025	ASTRO 100 / PHYSICS 107 Graduate Teaching Assistant	The University of Auckland

Student Supervision

Capstone (Undergraduate) Projects

2026	Ursula Brackstone-Dowd & Jaden Ferns Further Refinement of the Natal Supervisor Kick
2026	Nathan Adamson & Liam Venman Evaluating analytical approximations of Roche lobe overflow

Research Experience

University of Auckland - Department of Physics — Auckland, NZ
Supervisor: Assoc. Prof. Nicholas Rattenbury — June 2023 - December 2023
- Research Assistant: University of Auckland Ground Station One

University of Auckland - Department of Physics — Auckland, NZ
Supervisor: Prof. Jan (JJ) Eldridge — April 2021 - September 2021
- Research Assistant

University of Auckland - Department of Physics — Auckland, NZ
Supervisor: Assoc. Prof. Nicholas Rattenbury — December 2019 - February 2020
- Summer Research Scholar: "Ground Segment Support for APSS-I"
- This report is non-distributable due to a non-disclosure agreement

Outreach & Professional Development

Academic Citizenship and Service

2026-	Faculty of Science Equity Committee Member	The University of Auckland
2025	Department of Physics Equity Committee Chair	The University of Auckland
2023-2026	Department of Physics Equity Committee Member	The University of Auckland

Professional Memberships

2026-	Royal Astronomical Society Fellow	United Kingdom

Selected Media Appearances

2026	First Samoan astrophysicist encourages Pasifika students to reach for the stars Radio New Zealand

Outreach Activities

2021-2025	Manawa Mai Open Day Demonstrator	The University of Auckland
2021-2025	Future Me Demonstrator	The University of Auckland
2021-2025	Incredible Science Demonstrator	The University of Auckland

Societies

2021	Post-Graduate Students Association Welfare Vice President	The University of Auckland
2021	Physics Association of the University of Auckland Vice President	The University of Auckland
2020	Physics Association of the University of Auckland Treasurer	The University of Auckland
2018	Physics Association of the University of Auckland Committee Member	The University of Auckland

Professional Development

First Aid Certificate, September 2025 (expiry September 2027)

Publications

Published

Richards, S., Eldridge, J., Briel, M., Stevance, H., Willcox, R. 2023. *New constraints on the Bray conservation-of-momentum natal kick model from multiple distinct observations*. Monthly Notices of the Royal Astronomical Society.

Stevance, H., Ghodla, S., **Richards, S.**, Eldridge, J., Briel, M., Tang, P. 2023. *VFTS 243 as predicted by BPASS models*. Monthly Notices of the Royal Astronomical Society.

Richards S., Eldridge J., Briel M., Ghodla S. 2025. *The evolution of accretor stars in binary systems due to accretion of increasingly helium-rich material*. Monthly Notices of the Royal Astronomical Society.

In Review

In Prep

Mandhai S., Breton R., Kay S., **Richards, S.** 2026. *The orbital evolution of NSNS in Milky Way like galaxies*.

Richards, S., Eldridge, J., Bray, J., Gleave, J., Steed, M. 2026. *The Aotearoa STARS code*.

Richards, S. 2026. *Takahe: Compact object inspirals with BPASS*

Richards, S. 2026. *Kaitiaki: Watching over STARS*

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